THEORY OF PRODUCTION

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- Production = Creation of Utility
- Creation of Economic Utility

1. Form Utility
2. Place Utility
3. Time Utility
4. Service Utility

THEORY OF PRODUCTION

- Production involves transformation of inputs such as capital, equipment, labor, and land into output - goods and services

- In this production process, the manager is concerned with *efficiency* in the use of the inputs

  - technical vs. economical efficiency

Two Concepts of Efficiency

- Economic efficiency:
  - occurs when the cost of producing a given output is as low as possible
- Technological efficiency:
  - occurs when it is not possible to increase output without increasing inputs

You will see that basic production theory is simply an application of constrained optimization:

The firm attempts either to minimize the cost of producing a given level of output

or

to maximize the output attainable with a given level of cost.

Both optimization problems lead to same rule for the allocation of inputs and choice of technology

Production Function

- A *production function* is purely technical relation which connects factor inputs & outputs. It describes the transformation of factor inputs into outputs at any particular time period.

$$Q = f(L_{d,} L,K, E)$$

where

Q = output	R= Raw Material
L= Labour	L_d = Land
K= Capital	E = Entreprenure

For our current analysis, let's reduce the inputs to two, capital (K) and labor (L):

$$Q = f(L, K)$$

Production Table

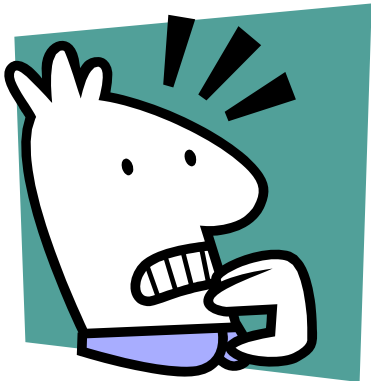
Units of K Employed	Output Quantity (Q)							
8	37	60	83	96	107	117	127	128
7	42	64	78	90	101	110	119	120
6	37	52	64	73	82	90	97	104
5	31	47	58	67	75	82	89	95
4	24	39	52	60	67	73	79	85
3	17	29	41	52	58	64	69	73
2	8	18	29	39	47	52	56	52
1	4	8	14	20	27	24	21	17
	1	2	3	4	5	6	7	8
	Units of L Employed							

Same Q can be produced with different combinations of inputs, e.g. inputs are substitutable in some degree

Short-Run and Long-Run Production

- In the short run some inputs are fixed and some variable
  - e.g. the firm may be able to vary the amount of labor, but cannot change the amount of capital
  - in the short run we can talk about *factor productivity* / ***Law of Return to a Factor/ Law of variable proportion / Law of diminishing returns***

- In the long run all inputs become variable
  - e.g. the long run is the period in which a firm can adjust *all* inputs to changed conditions
  - in the long run we can talk about ***Law of returns to scale/ Law of Constant Propotion***



Law of Variable Proportion/ Law of Return to a Factor

Fixed Factor (K)	Variable Factor (L)
1	2 (3hrs)
1	3 (6hrs)
1	4 (10 hrs)
1	5 (15 hrs)
1	**6 (18 hrs)** **Best Use of Machine**
1	7 (22 hrs)

Law of Constant Proportion/ Law of Returns to Scale

Capital (K)	Labour (L)
1	6
2	12
3	18
4	24

Short-Run Changes in Production

Factor Productivity

Units of K Employed	Output Quantity (Q)							
8	37	60	83	96	107	117	127	128
7	42	64	78	90	101	110	119	120
6	37	52	64	73	82	90	97	104
5	31	47	58	67	75	82	89	95
4	24	39	52	60	67	73	79	85
3	17	29	41	52	58	64	69	73
2	8	18	29	39	47	52	56	52
1	4	8	14	20	27	24	21	17
	1	2	3	4	5	6	7	8
	Units of L Employed							

How much does the quantity of Q change, when the quantity of L is increased?

Long-Run Changes in Production

Returns to Scale

Units of K Employed	Output Quantity (Q)							
8	37	60	83	96	107	117	127	128
7	42	64	78	90	101	110	119	120
6	37	52	64	73	82	90	97	104
5	31	47	58	67	75	82	89	95
4	24	39	52	60	67	73	79	85
3	17	29	41	52	58	64	69	73
2	8	18	29	39	47	52	56	52
1	4	8	14	20	27	24	21	17
	1	2	3	4	5	6	7	8
	Units of L Employed							

How much does the quantity of Q change, when the quantity of both L and K is increased?

Relationship Between Total, Average, and Marginal Product: Short-Run Analysis

- Total Product (TP) = total quantity of output
- Average Product (AP) = total product per total input
- Marginal Product (MP) = change in quantity when one additional unit of input used

The Marginal Product of Labor

- The marginal product of labor is the increase in output obtained by adding 1 unit of labor but holding constant the inputs of all other factors

Marginal Product of L:

$$MP_L = \Delta Q/\Delta L \quad \text{(holding K constant)}$$

$$= \delta Q/\delta L$$

Average Product of L:

$$AP_L = Q/L \quad \text{(holding K constant)}$$

Law of Diminishing Returns (Diminishing Marginal Product)

The law of diminishing returns states that ***when more and more units of a variable input are applied to a given quantity of fixed inputs****, the* ***total output may initially increase at an increasing rate and then increases at a decreasing rate and eventually starts falling****.*

Assumptions: The law of diminishing returns is based on the following assumptions:

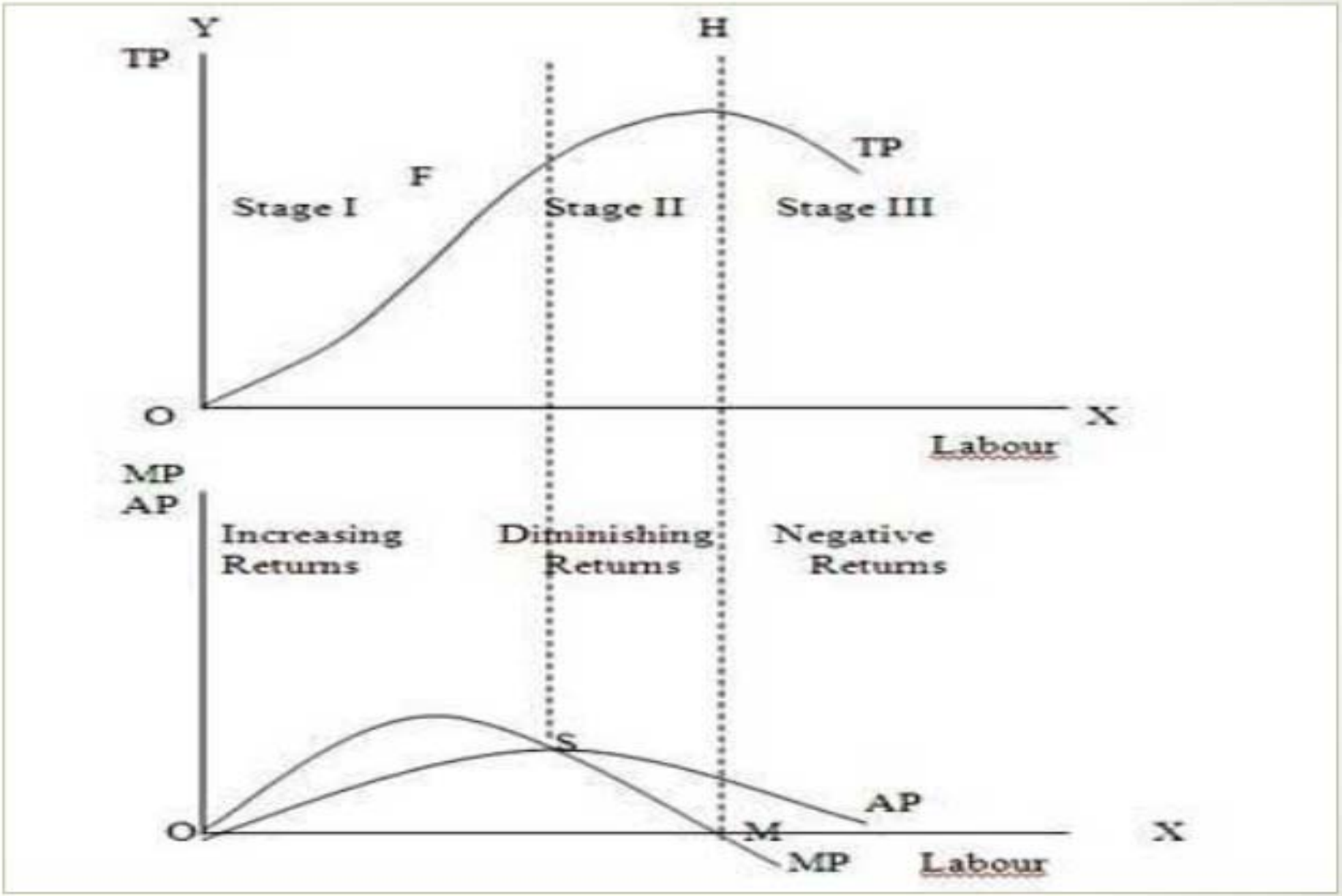
(i) the state of technology is given

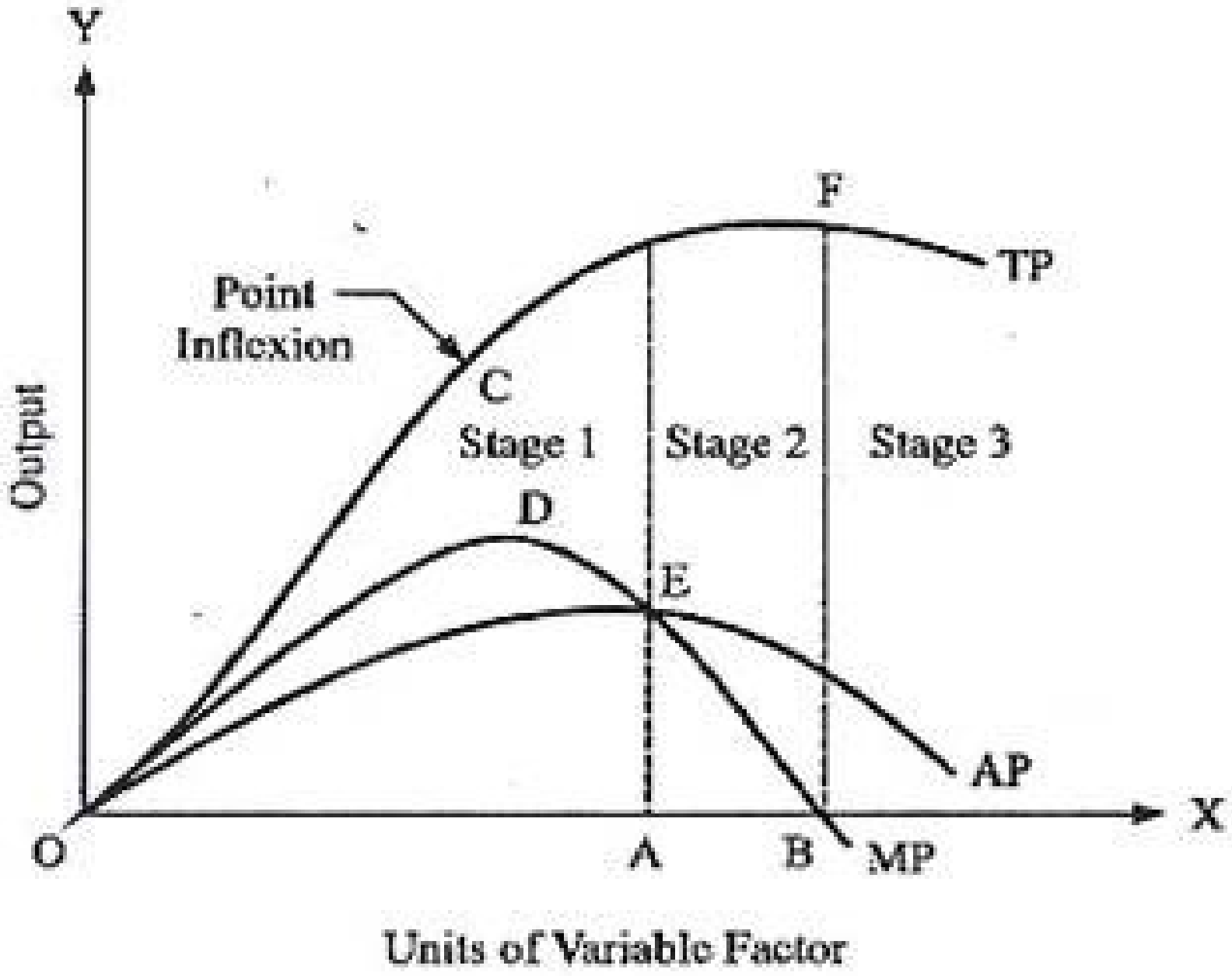
(ii) labour is homogenous and

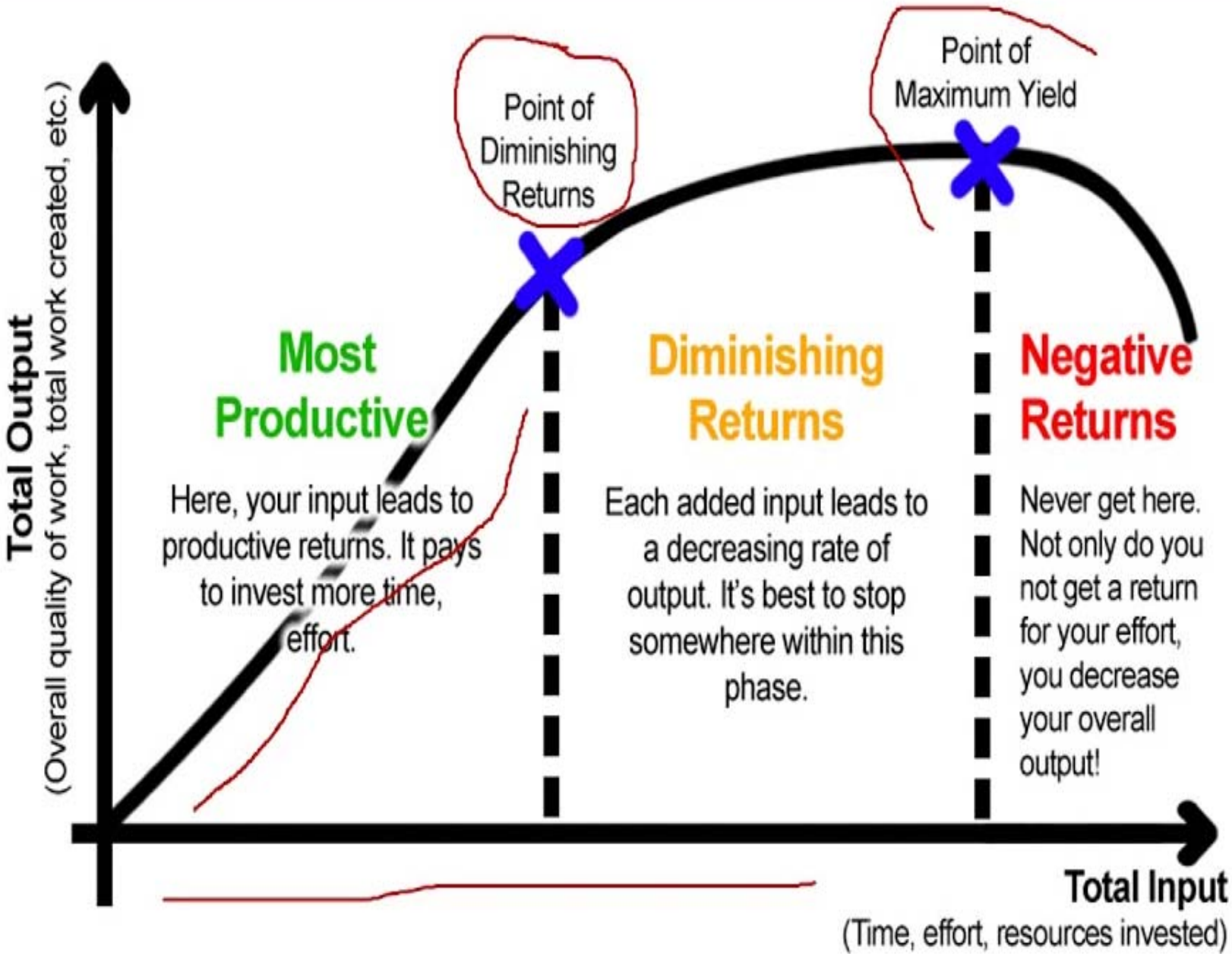
(iii) input prices are given

Labour (X)	Total Product (Q or TP)	Average Product (AP)	Marginal Product (MP)	Stages of Production
1	24	24	24	INCREASING RETURNS
2	72	36	48	
3	138	46	66	
4	216	54	78	
5	300	60	84	
6	**384**	**64**	**84**	
7	462	66	78	
8	**528**	**66**	**66**	
9	576	64	48	DIMINISHING RETURNS
10	600	60	24	
11	594	54	-6	NEGATIVE RETURNS
12	552	46	-42	

Short-Run Analysis of Total, Average, and Marginal Product




Y
F
TP
Point
Inflexion
C
Output
Stage 1
Stage 2
Stage 3
D
E
AP
X
O
A
B
MP
Units of Variable Factor


Point of Diminishing Returns
Point of Maximum Yield
Total Output
(Overall quality of work, total work created, etc.)
Most Productive
Here, your input leads to productive returns. It pays to invest more time, effort.
Diminishing Returns
Each added input leads to a decreasing rate of output. It's best to stop somewhere within this phase.
Negative Returns
Never get here. Not only do you not get a return for your effort, you decrease your overall output!
Total Input
(Time, effort, resources invested)

Application of Law of Diminishing Returns:

- It helps in identifying the rational and irrational stages of operations.
- It gives answers to question –

  How much to produce?

  What number of workers to apply to a given fixed inputs so that the output is maximum?

Production in the Long-Run

- All inputs are now considered to be variable (both L and K in our case)
- How to determine the optimal combination of inputs?

To illustrate this case we will use *production isoquants*.

An *isoquant* is a locus of all technically efficient methods or all possible combinations of inputs for producing a given level of output.

Production Table

Isoquant

Units of K Employed	Output Quantity (Q)							
8	37	60	83	96	107	117	[illegible]	128
7	42	64	78	90	101	110	119	120
6	37	52	64	73	82	90	97	104
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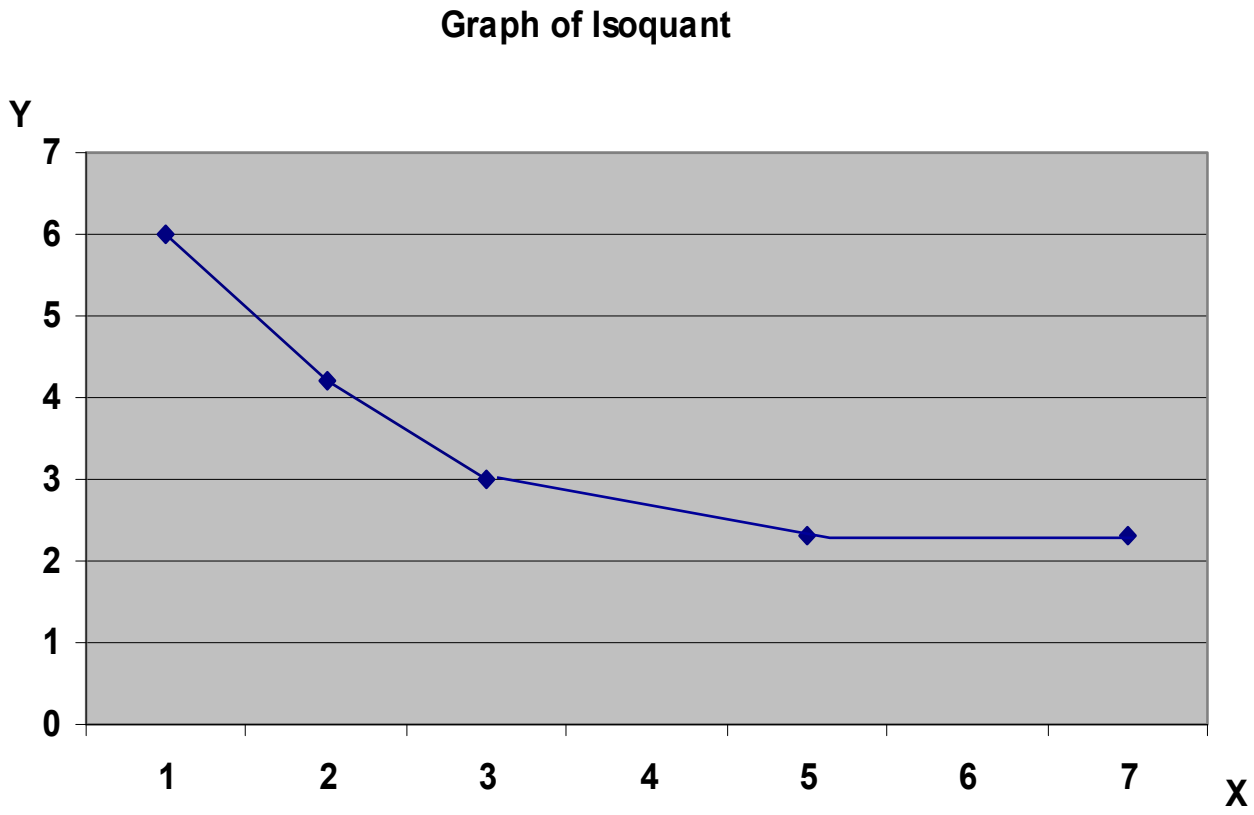
	1	2	3	4	5	6	7	8
	Units of L							

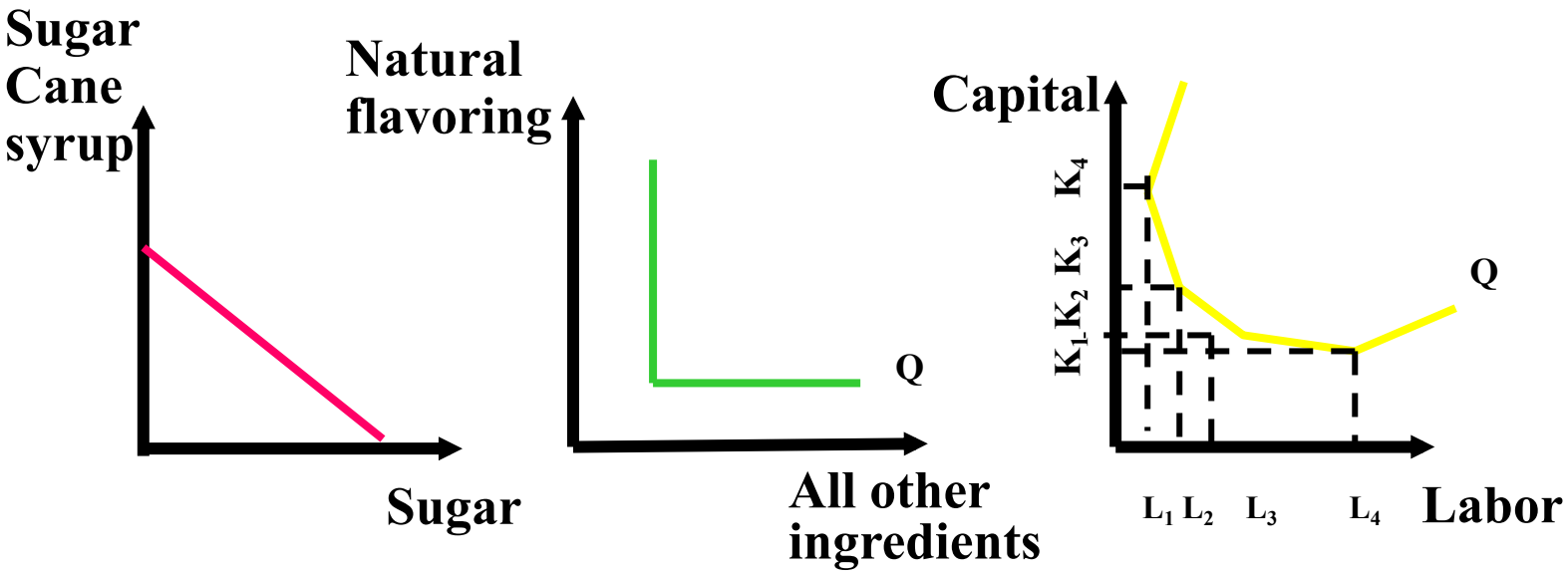
Isoquant



Types of Isoquant

There exists some degree of substitutability between inputs.

Different degrees of substitution:



a) Linear Isoquant (Perfect substitution)

b) Input – Output/ L-Shaped Isoquant (Perfect complementarity)

c) Kinked/Acitivity Analysis Isoquant – (Limited substitutability)

Marginal Rate of Technical Substitution MRTS

- The degree of imperfection in substitutability is measured with marginal rate of technical substitution (MRTS- Slope of Isoquant):

$$MRTS = \Delta L/\Delta K$$

(in this MRTS some of L is removed from the production and substituted by K to maintain the same level of output)

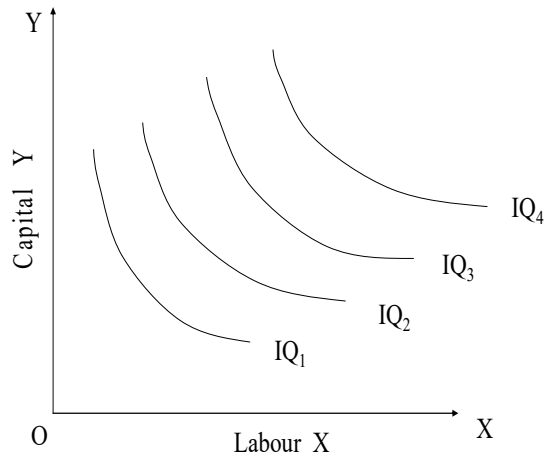
Properties of Isoquants

- Isoquants have a negative slope.
- Isoquants are convex to the origin.
- Isoquants cannot intersect or be tangent to each other.
- Upper Isoquants represents higher level of output

Isoquant Map

- Isoquant map is a set of isoquants presented on a two dimensional plain. Each isoquant shows various combinations of two inputs that can be used to produce a given level of output.

Figure : Isoquant Map



Laws of Returns to Scale

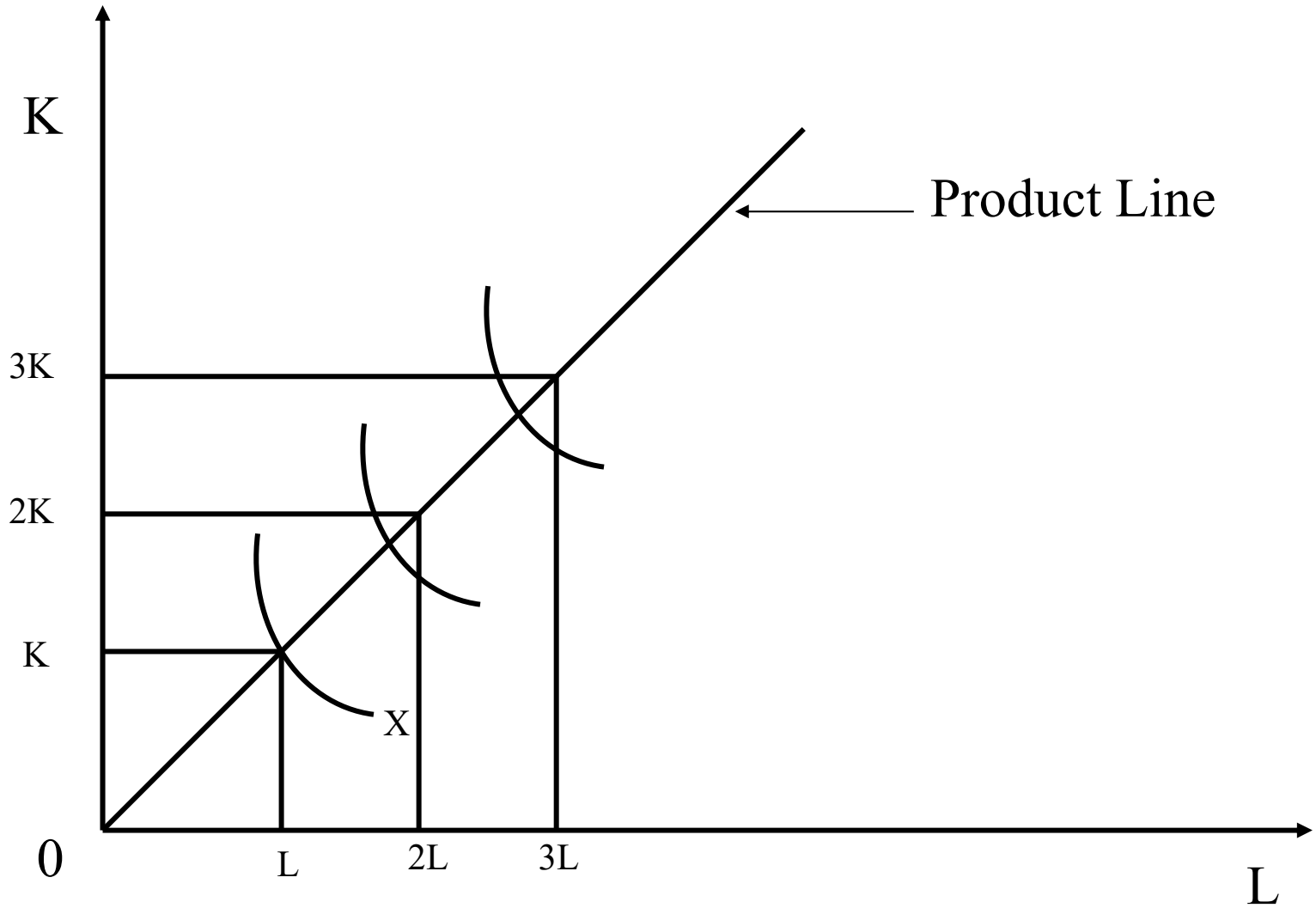
- It explains the behavior of output in response to a proportional and simultaneous change in input.
- When a firm increases both the inputs, there are three technical possibilities –

(i) TP may increase more than proportionately – Increasing RTS

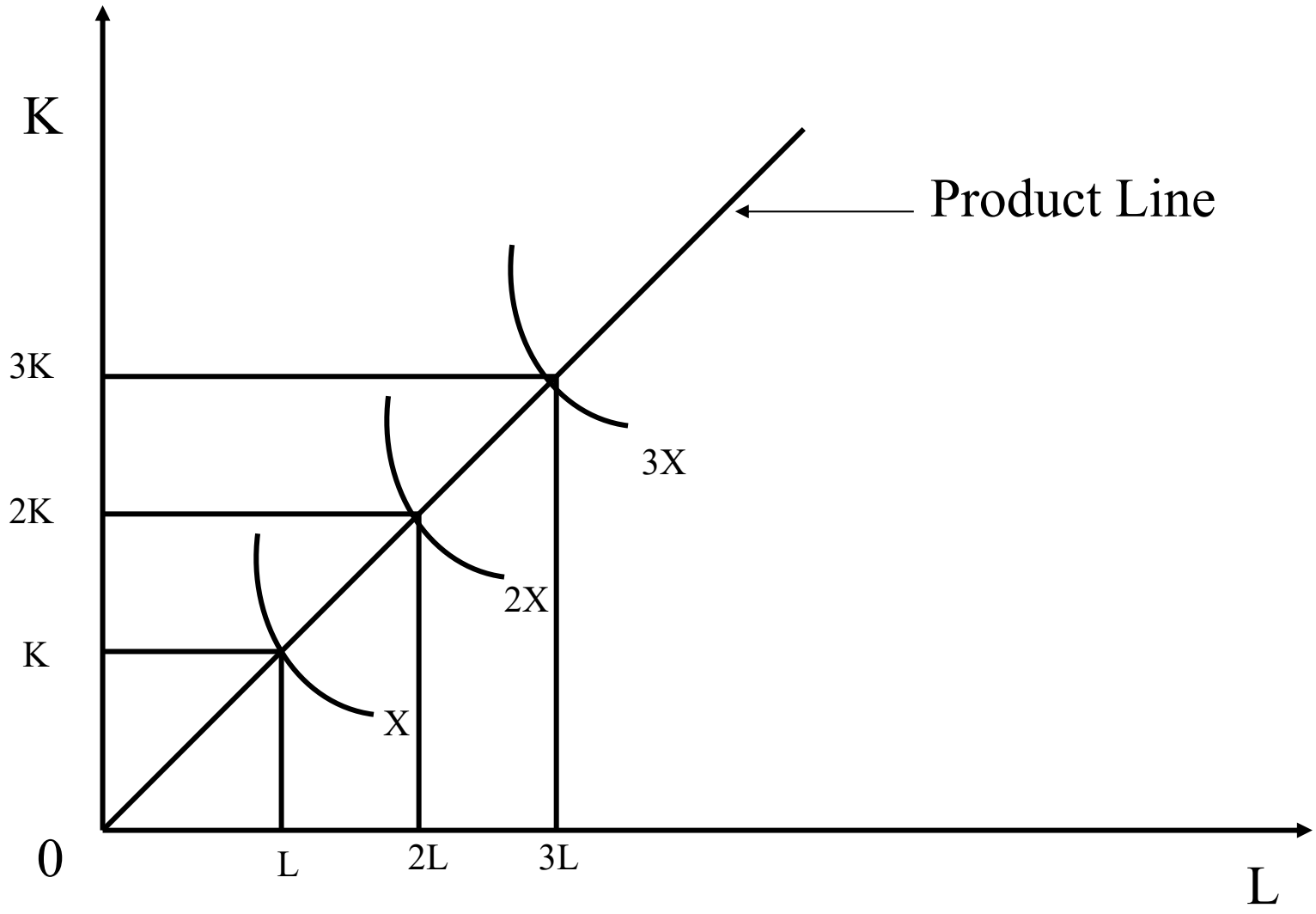
(ii) TP may increase proportionately – constant RTS

(iii) TP may increase less than proportionately – diminishing RTS

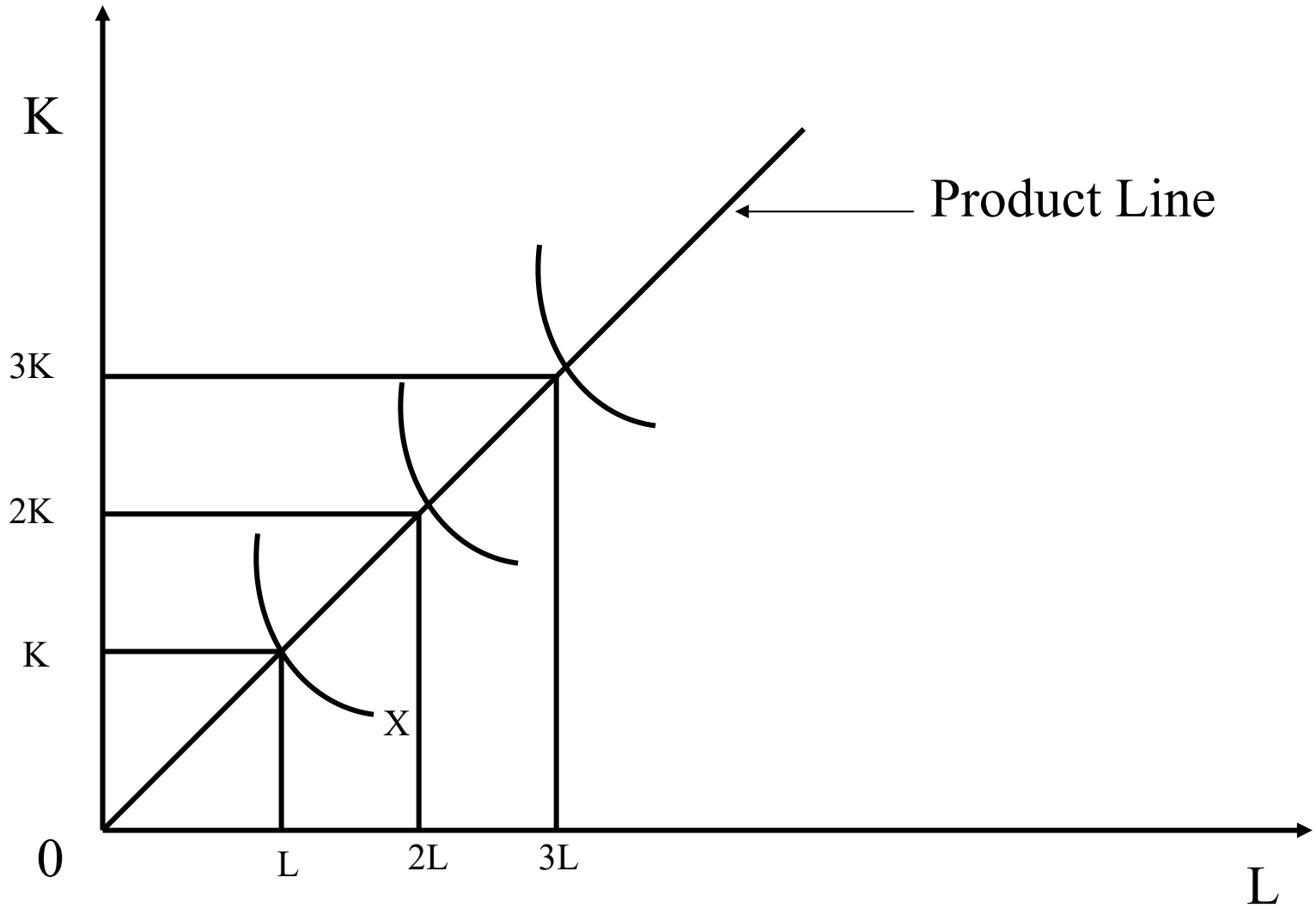
Increasing RTS



Constant RTS



Decreasing RTS



We will use isoquant map (1) and iso-cost line (2)

Figure : Isoquant Map (1)

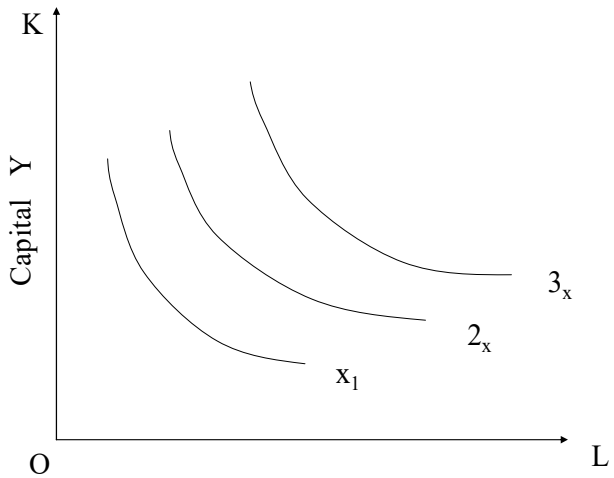
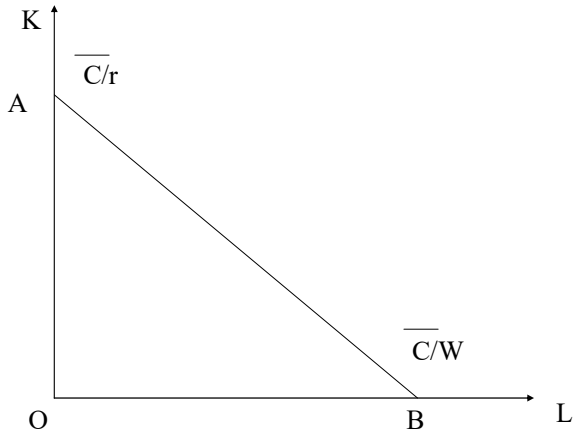


Figure : Isoquant Line (2)



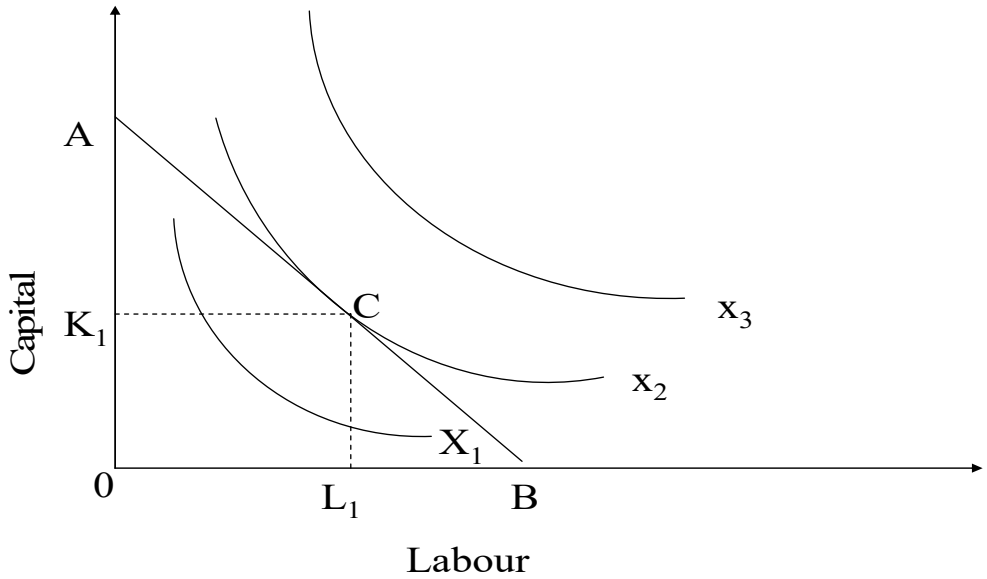
The cost line is defined by cost equation

C= (r) (k) + (w) (L)

W ⟶ wage rate r= price of capital service

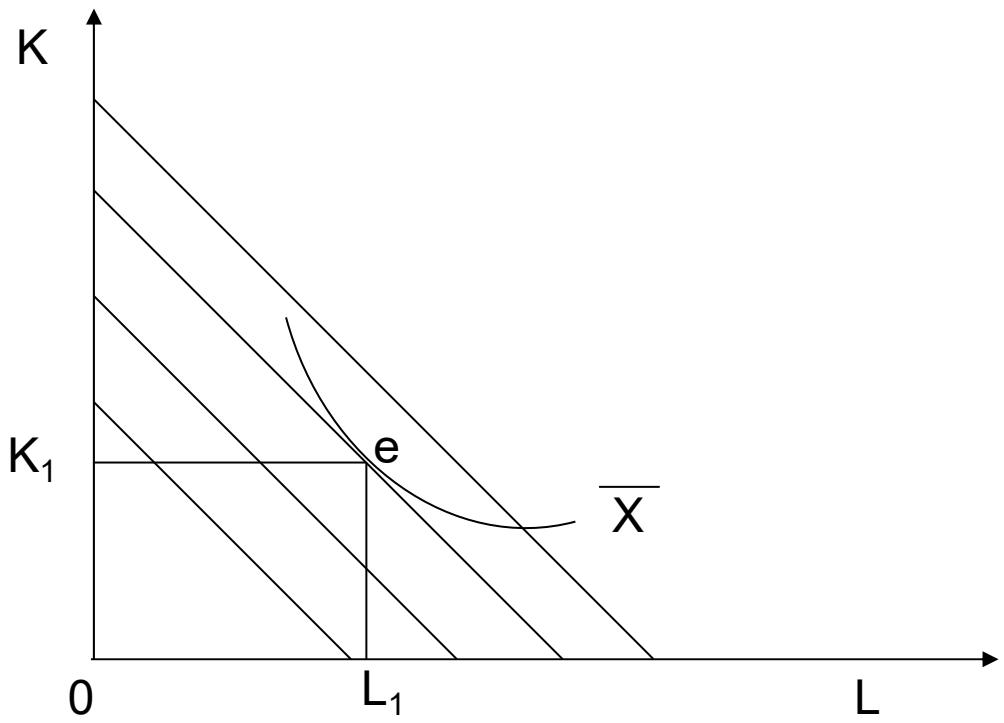
Case I

Maximization of output subject to cost constraint



Condition for Equilibrium

- At point of tendency slope of isocost line (w/r) = slope of isoquant. (MP_L/MP_K)
- The isoquants should be convex to origin

Case II

Minimization of cost for given level of output